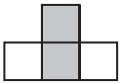
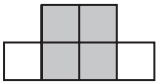
5 Here is a sequence of patterns made from grey squares and white squares.



pattern number 1

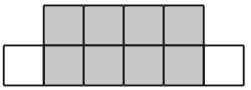


pattern number 2



pattern number 3

(a) In the space below, draw pattern number 4



1 mark
8 grey and 2 white squares

(1)

(b) Work out the total number of squares needed to make pattern number 7

Pattern n has 2n grey squares and 2 white squares.

Pattern 7: 2 × 7 + 2 = 16 squares.

2 marks total
Use 2n + 2
16 squares

(2)

Aqsa says,

"The total number of squares needed to make pattern number 20 is double the total number of squares needed to make pattern number 10"

(c) Is Aqsa correct?
Give a reason for your answer.

Pattern 10: 2 × 10 + 2 = 22 squares.

1 mark
Compare 22 and 42

Pattern 20: 2 × 20 + 2 = 42 squares.

No: double 22 is 44, not 42.

1 mark
No; 42 is not 44

(2)

(Total for Question 5 is 5 marks)